

# Shiyao Sang (Samuel)

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IEEE RAS Technical Committee on Cognitive Robotics — Member

**Research Focus:** Robot learning · Latent world models · Representation learning. My research focuses on planning-oriented representations for closed-loop autonomous systems, with recent work on task-driven sparse latent planning for autonomous driving and distributed robotic middleware for robot systems (first-author paper accepted at IROS 2025).

## Education

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|----------------------------------------------------------------------------------|------------------------------------------------|
| <b>University of Wollongong</b> , NSW, Australia                                 | <i>M.Sc., Computer Science</i>                 |
| Faculty of Engineering and Information Sciences                                  | 2020–2022                                      |
| Joint Dual-Degree Program with Central China Normal University                   |                                                |
| <b>Central China Normal University</b> , Wuhan, China                            | <i>M.Eng., Computer Technology</i>             |
| Faculty of Artificial Intelligence in Education                                  | 2020–2022                                      |
| Joint Dual-Degree Program with University of Wollongong                          |                                                |
| <b>Huaiyin Institute of Technology (now Huai'an University)</b> , Jiangsu, China | <i>B.Eng., Computer Science and Technology</i> |
| Faculty of Computer and Software Engineering                                     | 2014–2018                                      |

## Publications & Research Outputs

- **Shiyao Sang**, Shenghai Yuan, Yuzhi Lai, et al. “Task-Driven Sparse Latent Planning without Future World-State Supervision for Autonomous Driving.” *Manuscript submitted*, 2026.
- **Shiyao Sang**. “Discrete Gradient Policy Optimization via Grouped Trajectory Sampling for Structure-Agnostic Motion Planning.” *Manuscript in preparation*.
- **Shiyao Sang**, Yinggang Ling. “Service Discovery-Based Hybrid Network Middleware for Efficient Communication in Distributed Robotic Systems.” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. Video: <https://youtu.be/uMvyOykTlkA>
- **Shiyao Sang**. “Design and Implementation of a State Transition Model for Educational Robot Tutoring Math Homework.” *Master’s Thesis*, University of Wollongong & Central China Normal University, 2022.
- **Shiyao Sang**. “A Multi-Layer Architecture for Intelligent Chatbots in IoT Robots: Design and Implementation.” *Bachelor’s Thesis*, Huaiyin Institute of Technology, 2018.

## Professional Experience

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| <b>BYD Company Ltd. – New Technology Research Institute</b>                                                                                                                                                                                                                                                                                                                                        | Jun 2025 – Dec 2025 |
| <i>Senior Algorithm Engineer</i> , Urban Joint Game Group                                                                                                                                                                                                                                                                                                                                          |                     |
| <ul style="list-style-type: none"><li>• Designed and implemented joint game-theoretic, interaction-aware planning strategies for complex urban scenarios in BYD’s urban intelligent driving system.</li><li>• Conducted preliminary research on diffusion-generation-based game-theoretic planning, including baseline reproduction, technical scheme design, and prototype preparation.</li></ul> |                     |
| <b>Chery Automobile Co., Ltd. – Intelligentization Center</b>                                                                                                                                                                                                                                                                                                                                      | Mar 2023 – Mar 2025 |
| <i>Algorithm Engineer</i> , Foresight Algorithm Group                                                                                                                                                                                                                                                                                                                                              | May 2024 – Mar 2025 |
| <ul style="list-style-type: none"><li>• Worked on lateral decision-making lattice sampling and planning software architecture for an advanced autonomous driving planning framework.</li></ul>                                                                                                                                                                                                     |                     |

- Proposed a reinforcement-learning-based pruning search method for mapless intersection path generation, recognized as a Chery Group Class-A patent.

*Software Engineer*, Operating System Department

Mar 2023 – May 2024

- Designed and implemented communication middleware deployed on Robotaxi and Robotruck platforms; the related work was published as a first-author paper at IROS 2025.
- Designed a multi-Orin distributed computing architecture for Robotaxi, balancing algorithm deployment across computing resources and communication loads.

**Coowa Technology Co., Ltd.**

Jul 2022 – Jan 2023

*Software Engineer*, Remote Driving Group

- Developed network security and cockpit audio modules for an L4 autonomous vehicle teleoperation system; contributed an encryption-related fix MR to ZLMediaKit.

## Patents

- **Shiyao Sang.** “An Adaptive Trajectory Generation Method for Intersections Without High-Precision Maps Based on Multi-Deciders and Evaluators.” *Invention Patent*, 2024.
- **Shiyao Sang.** “SOME/IP Service to Topic Software Architecture Design Scheme.” *Invention Patent*, 2023.
- **Shiyao Sang.** “Digital Clock-based Automatic Latency Testing Method and System for Remote Driving Streaming Media.” *Invention Patent*, 2022.

## Technical Skills

- **AI & Learning:** Latent World Models, Deep Learning, Reinforcement Learning, Markov Decision Processes (MDP), Monte Carlo Tree Search (MCTS), and VLM-based Planning.
- **Robotic Middleware & Systems:** Expertise in distributed computing architectures (ROS, ROS2, CyberRT) and SOME/IP software architecture design.
- **Human-Robot Interaction (HRI):** Dialogue system design (AIML), prompt engineering, and LLM fine-tuning for interactive robotics.
- **Programming:** C++, C#, Python; PyTorch and TensorFlow.

## Honors & Awards

- 2017 “Zhongxing Cup” Pan-Pearl River Delta University Computer Works Competition — Silver (Undergraduate Group)
- 2017 China Engineering Robot Competition — IoT Robot Innovation — Second Prize
- 2018 Microsoft Imagine Cup — Suzhou Regional Final — Third Prize
- Jiangsu Province 5th Science Students Humanities & Social Science Knowledge Contest — Excellence Award

## Referees

Dr. Shenghai Yuan – Research Collaborator – Senior Research Fellow, Nanyang Technological University – [Shyuan@ntu.edu.sg](mailto:Shyuan@ntu.edu.sg)

Dr. Zhilong Liu – Former Technical Supervisor – Senior Algorithm Engineer, Woven by Toyota – [lzl200102109@gmail.com](mailto:lzl200102109@gmail.com)